Problem Set 2

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For every single question, unless otherwise noted I collaborated with Blair Moore, Thomas Li, and John Cho.

Problem 1

Problem 1.a.

For all $i \neq n$, the expectation is zero, $1/2 \cdot (-1) + 1/2 \cdot (1) = 0$. Note that for all realizations of ε_i, the squared value is exactly one. So $E[\varepsilon_i^2] - E[\varepsilon_i]^2 = E[1] - 0^2 = 1$. For covariance, with $i, j \neq n$, then by independence the covariance is zero.

For $i = n$. Note that the expectation:

$$\begin{aligned} E[\varepsilon_n] &= E[\prod_{i<n}(-1)\varepsilon_i] \\ &= \prod_{i<n}(-1)E[\varepsilon_i] && \text{(Independence)} \\ &= \prod_{i<n}(-1)0 \\ &= 0 \end{aligned}$$

Similarly to before, the only possible values of ε_n are -1 and 1, so the squares are only 1. The variance is:

$$\begin{aligned} Var(\varepsilon_n) &= E[\varepsilon_n^2] - E[\varepsilon_n]^2 \\ &= E[1] - 0^2 \\ &= 1 \end{aligned}$$

For the covariance between n and $i \neq n$, we do:

$$
\begin{aligned}
Cov(\varepsilon_n, \varepsilon_i) &= E[\varepsilon_n \cdot \varepsilon_i] - E[\varepsilon_i]E[\varepsilon_n] \\
&= E[\varepsilon_n \cdot \varepsilon_i] - 0 \cdot 0 \\
&= E[\prod_{j<n}(-1)\varepsilon_i \cdot \varepsilon_i] \\
&= E[(-\varepsilon_i)(\varepsilon_i) \prod_{j<n, j\neq i}(-1)\varepsilon_i \cdot \varepsilon_i] \\
&= E[(-\varepsilon_i)(\varepsilon_i)] \cdot E[\prod_{j<n, j\neq i}(-1)\varepsilon_i \cdot \varepsilon_i] && \text{(Independence)} \\
&= E[(-\varepsilon_i)(\varepsilon_i)] \cdot \prod_{j<n, j\neq i}(-1)E[\varepsilon_i \cdot \varepsilon_i] && \text{(Independence)} \\
&= E[1] \cdot \prod_{j<n, j\neq i}(-1)0 \\
&= 1 \cdot 0 \\
&= 0
\end{aligned}
$$

Problem 1.b.

We can compute the expected value with

$$
\begin{aligned}
E[\hat{\beta}] &= E\left[\frac{1}{n}\sum y_i\right] \\
&= \frac{1}{n}E\left[\sum y_i\right] \\
&= \frac{1}{n}\sum E[y_i] \\
&= \frac{1}{n}\sum E[\beta_0 + \varepsilon_i] \\
&= \frac{1}{n}\sum E[\beta_0] + E[\varepsilon_i] \\
&= \frac{1}{n}\sum \beta_0 + \frac{1}{n}\sum E[\varepsilon_i] \\
&= \frac{1}{n}n\beta_0 + \frac{1}{n}\sum E[\varepsilon_i] \\
&= \frac{1}{n}n\beta_0 + \frac{1}{n}\sum 0 \\
&= \frac{1}{n}n\beta_0 + 0 \\
&= \beta_0
\end{aligned}
$$

We can compute the variance by noting that $\hat{\beta} = \frac{1}{n}\mathbf{1}^T\mathbf{y}$. So its variance will be: $\hat{\beta} = \frac{1}{n^2}\mathbf{1}^T\Sigma\mathbf{1}$. But as argued previously, the off-diagonal elements are all zero and all of the on-diagonal elements are exactly one. So the variance must be $\frac{1}{n^2}\mathbf{1}^T I\mathbf{1} = \frac{n}{n^2} = \frac{1}{n}$.

Problem 1.c.

Since there are no covariates, a linear estimator will always be of the form: $a^T y$ For a vector of weights, a and the vector of data y.

Unbiasedness requires:

$$
\begin{aligned}
\beta_0 &= E[a^T y] \\
\beta_0 &= E[a^T(\beta_0 1 + \varepsilon)] \\
\beta_0 &= E[\beta_0 a^T 1] + E[a^T \varepsilon] \\
\beta_0 &= \beta_0 E[a^T 1] + a^T E[\varepsilon] \\
\beta_0 &= \beta_0 E[a^T 1] + a^T 0 \\
\beta_0 &= \beta_0 E[a^T 1] \\
1 &= E[a^T 1]
\end{aligned}
$$

This gives a simple constraint for unbiasedness of the linear estimator.

The variance of such an estimator will be:

$$
\begin{aligned}
Var(a^T y) &= Var(a^T(\beta_0 1 + \varepsilon)) \\
&= Var(\beta_0 a^T 1 + a^T \varepsilon) \\
&= Var(a^T \varepsilon) \\
&= a^T I a \\
&= a^T a
\end{aligned}
$$

So the minimal variance linear unbiased estimator solves the following Langrian:

$$
\mathcal{L} : a^T a - \lambda(a^T 1 - 1)
$$

With first order conditions:

$$
\begin{aligned}
2a - \lambda 1 = 0 &\implies a = \lambda/2 \cdot 1 \implies a = 1/n \cdot 1 \\
a^T 1 = 1 &\implies \lambda/2 \cdot 1^T \cdot 1 = 1 \implies \lambda = 2/n
\end{aligned}
$$

Therefore the minimum variance estimator is exactly the one that puts each weight on each value of y, exactly what $\hat{\beta}$ does.

Problem 1.d.

Note that there are only two possible values of y, $\beta_0 + 1$ or $\beta_0 - 1$. Clearly, if the value $y = \beta_0 + 1$ ever occurs, then $\max\{y\} = \beta_0 + 1$. So the maximum minus one is β_0. Now I will argue that $\beta_0 + 1$ always occurs. Suppose that any of the first $n - 1$ have an $\varepsilon_i = 1$, then we have the desired value, so we will correctly recover β_0. If the first $n - 1$ values are -1, then by assumption $\varepsilon_n = \prod -(-1) = \prod 1 = 1$.

So the maximum will always be $\beta_0 + 1$, so we always correctly recover β_0. Since there is only one possible value, the variance must also be zero.

Problem 1.e.

The maximum is not a linear operator, so it is allowed to have lower variance while being unbiased without violating the gauss markov theorem.

Problem 2

Problem 2.a.

I want to be precise about what this question is asking because I think there's some things that are implicit but require explication to get a rigorous answer.

In particular, I will **define** $\tilde{\beta}$ as follows:

$$(\tilde{\alpha}, \tilde{\beta}) \equiv \arg\min_{a,b} E[(E[y|x] - (a + bx))^2]$$

But since we know the true DGP, that simplifies to:

$$(\tilde{\alpha}, \tilde{\beta}) \equiv \arg\min_{a,b} E[\big(E[\alpha + \beta x + \gamma x^2 + \varepsilon] - (a + bx)\big)^2]$$

By linearity of expectations and conditional mean zero, we can further simplify to:

$$\begin{aligned}
(\tilde{\alpha}, \tilde{\beta}) &\equiv \arg\min_{a,b} E[\big(E[\alpha + \beta x + \gamma x^2 + \varepsilon|x] - (a + bx)\big)^2] \\
&= \arg\min_{a,b} E[\big(\alpha + \beta x + \gamma x^2 + E[\varepsilon|x] - (a + bx)\big)^2] \\
&= \arg\min_{a,b} E[\big(\alpha + \beta x + \gamma x^2 + 0 - (a + bx)\big)^2] \\
&= \arg\min_{a,b} E[\big(\alpha + \beta x + \gamma x^2 - (a + bx)\big)^2]
\end{aligned}$$

Since the problem is globally convex, the optimal parameters are pinned down with the first order conditions:

$$\begin{aligned}
\frac{d}{da} &: E[\alpha + \beta x + \gamma x^2 - (a + bx)] &= 0 \\
\frac{d}{db} &: E[\big(\alpha + \beta x + \gamma x^2 - (a + bx)\big)(x)] &= 0
\end{aligned}$$

The second first order condition is something like a covariance, so we can use the covariance decomposition, $Cov(x, y) = E[xy] - E[x]E[y]$ to rewrite the first order condition as:

$$
\begin{aligned}
0 &= E[\big(\alpha + \beta x + \gamma x^2 - (a + bx)\big)\,(x)] \\
&= Cov(\alpha + \beta x + \gamma x^2 - (a + bx), x) + \underbrace{E[\alpha + \beta x + \gamma x^2 - (a + bx)]}_{\text{0 by first FOC}} E[x] \\
&= Cov(\alpha + \beta x + \gamma x^2 - (a + bx), x) + 0E[x] \\
&= Cov(\alpha + \beta x + \gamma x^2 - (a + bx), x) \\
&= Cov((\alpha - a) + (\beta - b)x + \gamma x^2, x) \\
&= Cov(\beta - b)x + \gamma x^2, x) \\
&= Cov((\beta - b)x, x) + Cov(\gamma x^2, x) \\
&= (\beta - b)Cov(x, x) + \gamma Cov(x^2, x) \\
&= (\beta - b)Var(x) + \gamma Cov(x^2, x) \\
&= \beta Var(x) - bVar(x) + \gamma Cov(x^2, x) \\
b &= \frac{\beta Var(x)}{Var(x)} + \frac{\gamma Cov(x^2, x)}{Var(x)} \\
\tilde{\beta} &= \beta + \gamma \frac{Cov(x^2, x)}{Var(x)}
\end{aligned}
$$

Finally recovering the familiar omitted variable bias form.

But since we're looking at the covariance between a random variable and a function of itself, we can simplify further to give:

$$
\begin{aligned}
\tilde{\beta} &= \beta + \gamma \frac{Cov(x^2, x)}{Var(x)} \\
&= \beta + \gamma \frac{E[x^2 x] - E[x^2]E[x]}{Var(x)} \\
&= \beta + \gamma \frac{E[x^3] - (Var(x) + E[x]^2)\mu}{Var(x)} \\
&= \beta + \gamma \frac{E[x^3] - (1 + \mu^2)\mu}{1} \\
&= \beta + \gamma \left(E[x^3] - (1 + \mu^2)\mu\right)
\end{aligned}
$$

Where we are told that the variance is 1 and standard deviation is μ. In this part of the question, we assume that $\mu = 0$, so the thing being subtracted is zero. Furthermore, since the mean is zero, and the normal distribution is symmetric, all odd powers of the mean-zero normal will stay as symmetric distributions, so the third moment of x is also zero. Meaning that in this case, we have exactly $\tilde{\beta} = \beta$.

Now we should be precise in what that means, that means in the case where the xs are mean zero and symmetric, the $\tilde{\beta}$ we are trying to estimate is exactly the same as the β from the full data generating process. Since we know that the $\hat{\beta}$ estimating $\tilde{\beta}$ is unbiased (for all functional forms under the best linear predictor framework), we know that the expectation of $\hat{\beta}$ from the short regression is the same as the slope coefficient, β in the true model.

Problem 2.b.

In the case of $\mu \neq 0$, we cannot apply the same tricks we did above. In particular, the third moment will no longer be zero. However, since the family of normal distributions are closed under constant addition, we can represent x as $z + \mu$ where z is a standard normal distribution.

Then we can compute $E[x^3] = E[(z + \mu)^3]$ with:

$$
\begin{aligned}
E[x^3] &= E[(z+\mu)^3] \\
&= E[z^3 + 3z^2\mu + 3z\mu^2 + \mu^3] \\
&= E[z^3] + 3\mu E[z^2] + 3\mu^2 E[z] + \mu^3 \\
&= 0 + 3\mu(Var(x) + E[z]^2) + 3\mu^2 \cdot 0 + \mu^3 \\
&= 3\mu(Var(x) + E[z]^2) + \mu^3 \\
&= 3\mu(Var(x) + 0^2) + \mu^3 \\
&= 3\mu(Var(x)) + \mu^3 \\
&= 3\mu(1) + \mu^3 \\
&= (3+\mu^2)\mu
\end{aligned}
$$

From before we have:

$$
\begin{aligned}
\tilde{\beta} &= \beta + \gamma\left(E[x^3] - (1+\mu^2)\mu\right) \\
&= \beta + \gamma\left((3+\mu^2)\mu - (1+\mu^2)\mu\right) \\
&= \beta + \gamma\mu\left(3 + \mu^2 - 1 - \mu^2\right) \\
&= \beta + 2\gamma\mu
\end{aligned}
$$

Problem 3

Problem 3.a.

The "correctly" specified model is estimated below:

```
data <- read.csv('nls.csv')
baseline_lm <- lm(luwe ~ 1 + exper + educ, data = data)

X <- cbind(`(Intercept)` = rep(1,nrow(data)),exper = data$exper, educ = data$educ)
y <- data$luwe

beta_hat <- solve(t(X)%*% X)%*% t(X)%*% y

# sanity check
stopifnot(all.equal(setNames(c(beta_hat),rownames(beta_hat)), baseline_lm$coefficients))


print(beta_hat)
```

```
##                   [,1]
## (Intercept) 4.39731821
## exper       0.02352861
## educ        0.09111207
```

Comparing this to a model without `educ` gives the following:

```
X_short <- X[,-3]

beta_hat_short <- solve(t(X_short)%*% X_short)%*% t(X_short)%*% y


print(beta_hat_short)
```

```
##                      [,1]
## (Intercept)  6.042329594
## exper       -0.007115058
```

We can explain the change in sign `exper` using the omitted variable bias formula:

```
var_ratio <- cov(X[,'exper'],X[,'educ'])/var(X[,'exper'])
ovb <- beta_hat['exper',] + (beta_hat['educ',]*var_ratio)

print("Can we compute bias in short regression?")
```

```
## [1] "Can we compute bias in short regression?"
```

```
all.equal(unname(ovb), unname(beta_hat_short['exper',]))
```

```
## [1] TRUE
```

Problem 3.b.

```
H_hat <- (1/nrow(X))*t(X)%*%X
residuals <- y-X%*%beta_hat
sigma_squared <- ((t(residuals) %*% residuals)/length(residuals))[1,1]

Sigma_homoskedastic <-  sigma_squared * solve(H_hat) / length(y)
homosked_ses <- sqrt(diag(Sigma_homoskedastic))

cat("Homoskedastic Standard Errors:")
```

```
## Homoskedastic Standard Errors:
```

```
print(homosked_ses)
```

```
## (Intercept)       exper        educ
## 0.145395995 0.004345491 0.007564844
```

```
stopifnot(all(abs(Sigma_homoskedastic-vcov(baseline_lm))<.01))
```

Problem 3.c.

```
J_hat = (1/nrow(X))*t(X)%*%diag(c(residuals^2)) %*%X

Sigma_heteroskedastic <- solve(H_hat) %*% J_hat %*% solve(H_hat) / length(y)
heteroskeastic_ses <- sqrt(diag(Sigma_heteroskedastic))

print("Heteroskedastic Standard Errors:")
```

```
## [1] "Heteroskedastic Standard Errors:"
```

```
print(heteroskeastic_ses)
```

```
## (Intercept)       exper        educ
## 0.147698019 0.004370914 0.007705437
```